AudioConditioner

Results & Applications

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CSC_52002_EP - Multimodal generative AI

Application I: Image to Ambient Music

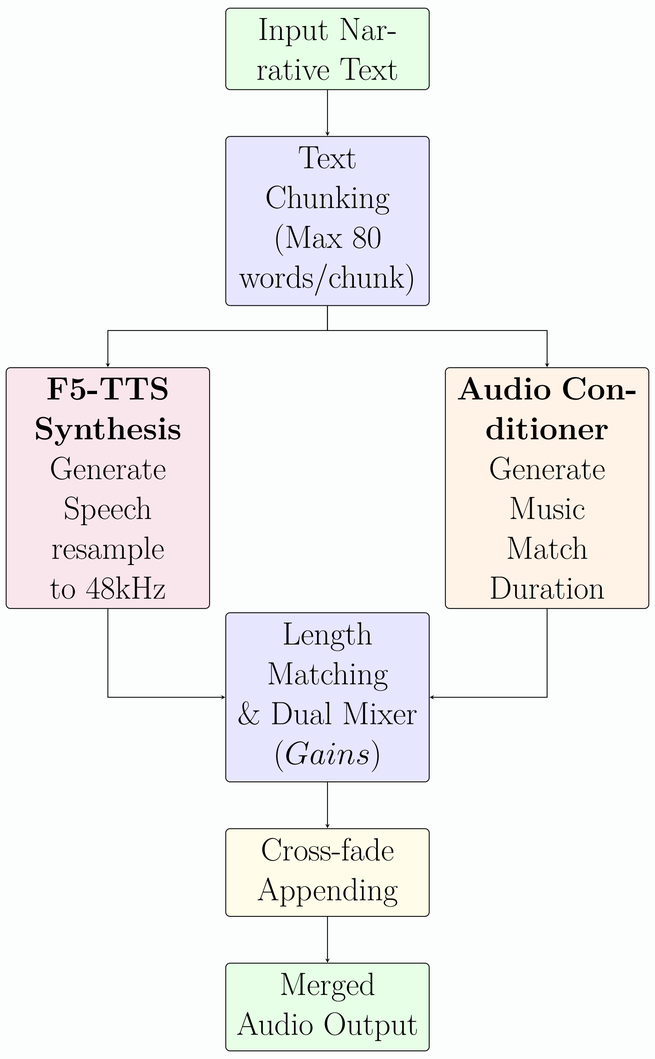
One direct application of the **AudioConditioner** framework is generating ambient music and atmospheres based on visual inputs. By passing an image into our pipeline:

- We use **BLIP** to automatically focus on the scene, extracting elements, colors, and objects.
- This scene description drives the AudioConditioner pipeline (through the CLAP Descriptor and Stable Audio); we aim to match the mood of the image in the generated audio.
- We re-rank the audio candidates using the CLAP cosine similarity filter to prioritize alignment with the initial visual.

This enables automated soundtrack generation for static artwork, game environments, or visual media.

Application II: Text Reading with Adaptive Music

The **ReadMusic** module combines speech synthesis with adaptive background music scoring to produce narrated audiobooks with adaptive background music.



We split the text using natural pauses such as punctuation or newlines. For each chunk (required by the 47-second generation maximum of the Stable Audio model), we run the **F5-TTS** engine concurrently with our music generation model. We zero-pad the speech and music arrays to match lengths, multiply them by specific gains, and blend them with dynamic crossfades.

Results: Semantic Alignment Benchmark

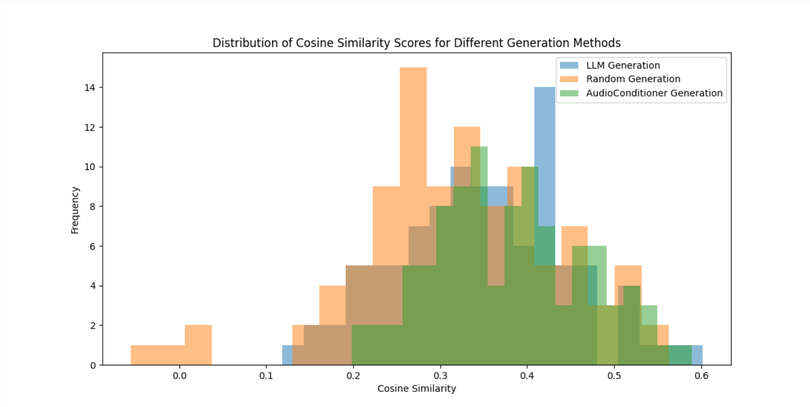


Figure 1: CLAP Similarity Distribution (via `benchmark.py`)

Method	Mean CLAP Similarity
AudioConditioner	0.38
LLM Baseline (Qwen2.5-7B)	0.35
Random Baseline	0.15

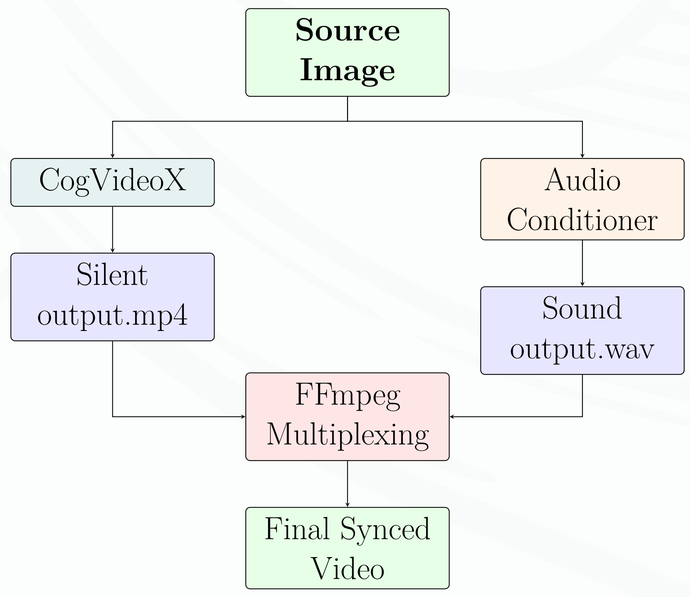
We benchmarked semantic alignment using CLAP cosine similarity between input text and generated audio across 100 held-out samples. AudioConditioner achieved a mean similarity of 0.38, compared to 0.35 for the direct LLM baseline.

Application III: Video Generation & Orchestration

We developed the `video_sound.py` application to create animated cinematic clips with a synchronized musical score from a single still image.

Parallel Execution Pathway:

1. **Image Animation:** We parse the input image through the **CogVideoX** model. As we strictly utilized an open-weight model, the resulting video quality remains limited but serves as a proof-of-concept for the pipeline.
2. **Image Orchestration:** Simultaneously, we pass the image directly into the **AudioConditioner** model to extract visual semantics and produce a matching audio waveform.
3. **Multimodal Merging:** We integrate **FFmpeg** to compile the generated silent video and the generated audio, multiplexing them into a final synced file.



These applications demonstrate flexibility across three pipelines for automated scoring, procedural art generation, and multimodal storytelling. We note that the quality of the generated video is lower than the audio quality, with visible artifacts.

Conclusion & Future Perspectives

We demonstrate that **AudioConditioner** steers music diffusion via a lightweight descriptor bridge trained by knowledge distillation. We achieved three working pipelines — image-to-music, text-to-narrated-audiobook, and image-to-video-with-score — with measurable semantic alignment.

Future work focuses on contrastive fine-tuning using **Reinforcement Learning**. By utilizing **Proximal Policy Optimization (PPO)** with the CLAP similarity score as a dense reward signal, we aim to optimize the descriptor weights to maximize semantic alignment directly. Given that a single audio generation step requires approximately 1 minute, a full PPO training cycle would require several days of continuous execution; however, this would allow the model to discover musical nuances that distillation from a teacher LLM alone may not capture. Longer-term trajectories include human perceptual evaluation and expanded training datasets to further refine the descriptor's precision.

References

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- Elizalde et al. (2023). *CLAP*: Learning Audio Concepts From Natural Language Supervision.
- Evans et al. (2024). *Stable Audio Open*: Latent Diffusion for Music Generation.
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